

# CSCI 131 — Introduction to Programming in Python

## Programming Assignment 6

### Loop Patterns and Lists

Open IDLE, choose **File** → **New File** for each program, and save it under the name given in the problem. Put your name in a comment on the first line. Where a problem gives you a list, type it into your program exactly as shown.

## Problem 1 — Class Roster

Write a program `roster.py` which starts with this list of students:

```
roster = ['Ana', 'Falilou', 'Khailah', 'Jadon', 'Afia', 'Tyson']
```

The program asks for a name and prints whether that name is on the roster. Use the prompt `name = input('Name: ')`. Names must match exactly, so `ana` is not the same as `Ana`.

Two sample runs:

```
Name: Ana
Ana is on the roster.
```

```
Name: Zoe
Zoe is not on the roster.
```

Your program must print exactly one line, no matter how long the roster is.

## Problem 2 — Temperatures

Write a program `temps.py` which starts with this list of daily high temperatures for two weeks:

```
highs = [78, 84, 81, 69, 90, 88, 75, 31, 45, 82, 79, 93, 66, 58]
```

The program prints how many days were *over* 80, the hottest temperature, and whether any day was below freezing (below 32). For the list above, it prints:

```
Days over 80: 6
Hottest: 93
There was a day below freezing.
```

If no day was below freezing, the last line is `No days below freezing.` instead.

## Submission

Submit your source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.